

The potential set of the light cone

A remark for the Mathematical Theory Atlas

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Status. Definitions and Propositions 2–3 are proved here from the explicit formula and Weil’s criterion; the converse in Proposition 2(iii) is Krein’s extension theorem and is cited; Computation 4 is taken from [5, Table 1]; Remarks make no claim.

Setup

Let W be the Weil distribution on $\mathbb{R}$, so that for every even $g \in C_c^\infty(\mathbb{R})$ the explicit formula reads

$$\sum_{\rho} \widehat{g}(\gamma_{\rho}) = \langle W, g \rangle, \quad \langle W, g \rangle = \widehat{g}\left(\frac{i}{2}\right) + \widehat{g}\left(-\frac{i}{2}\right) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) + W_{\infty}(g), \quad (1)$$

where $\rho = \frac{1}{2} + i\gamma_{\rho}$ runs over the nontrivial zeros with multiplicity, $\gamma_{\rho} \in \mathbb{C}$, and W_{∞} is the archimedean term [1, 5]. For g supported in $(-2a, 2a)$ only the prime powers with $\log n < 2a$ enter: the light cone of [5, §12]. Write $T^*(a) = 2\pi e^{2a}$ for its horizon.

Definition 1 (The potential set). For $a > 0$ let $\mathcal{P}(a)$ be the set of positive Radon measures ν on $\mathbb{R}$, symmetric under $t \mapsto -t$ and with $\int (1 + t^2)^{-1} d\nu < \infty$, such that

$$\int_{\mathbb{R}} \widehat{g} d\nu = \langle W, g \rangle \quad \text{for every even } g \in C_c^\infty(-2a, 2a). \quad (2)$$

An element of $\mathcal{P}(a)$ is a configuration of real “zeros” that no prime power inside the cone can distinguish from the true zeros. Put $\nu_{\zeta} = \sum_{\rho} \delta_{\gamma_{\rho}}$ when all γ_{ρ} are real.

Proposition 2. (i) $\mathcal{P}(a)$ is nonincreasing in a : $a < a'$ implies $\mathcal{P}(a') \subseteq \mathcal{P}(a)$.

(ii) If $\mathcal{P}(a) \neq \emptyset$ then the Weil form is nonnegative at support a : $\langle W, f * \tilde{f} \rangle \geq 0$ for every $f \in C_c^\infty(-a, a)$, in both the even and the odd sector.

(iii) Conversely, if $\langle W, f * \tilde{f} \rangle \geq 0$ for every $f \in C_c^\infty(-a, a)$, then $\mathcal{P}(a) \neq \emptyset$ (Krein’s extension theorem [3], in its distributional form [4]).

Proof. (i) The condition (2) for a' contains the condition for a . (ii) For $f \in C_c^\infty(-a, a)$ the function $g = f * \tilde{f}$ is even, supported in $(-2a, 2a)$, and $\widehat{g} = |\widehat{f}|^2 \geq 0$; hence $\langle W, g \rangle = \int |\widehat{f}|^2 d\nu \geq 0$ for any $\nu \in \mathcal{P}(a)$. (iii) The restriction of W to $(-2a, 2a)$ is a positive-definite distribution. Krein’s theorem produces a positive-definite distribution on $\mathbb{R}$ agreeing with it on $(-2a, 2a)$; by the Bochner–Schwartz theorem that distribution is the Fourier transform of a positive tempered measure ν , which is symmetric because W is even, and (2) holds by construction.¹ $\square$

¹Krein’s original statement concerns continuous positive-definite functions on an interval. The operator-theoretic proof (self-adjoint extensions of the generator of translations on the Hilbert space of the form) uses only the Hilbert-space structure and translation invariance and applies to positive-definite distributions; see [4]. The reader who prefers not to rely on this may take (iii) as the definition of the regime in which $\mathcal{P}(a)$ is studied.

Proposition 3 (The limit of the potential set).

$$\bigcap_{a>0} \mathcal{P}(a) = \begin{cases} \{\nu_\zeta\} & \text{if the Riemann Hypothesis holds,} \\ \emptyset & \text{otherwise.} \end{cases}$$

Proof. Let $\nu \in \bigcap_a \mathcal{P}(a)$. Then (2) holds for every even $g \in C_c^\infty(\mathbb{R})$, and by (1)

$$\int_{\mathbb{R}} \hat{g} d\nu = \sum_{\rho} \hat{g}(\gamma_{\rho}) \quad \text{for all even } g \in C_c^\infty(\mathbb{R}). \quad (3)$$

If all γ_{ρ} are real, the right side is $\int \hat{g} d\nu_{\zeta}$, and since the functions $\hat{g}$ are dense in the Schwartz space, the two tempered measures ν and ν_{ζ} coincide; conversely $\nu_{\zeta} \in \mathcal{P}(a)$ for every a by (1). If some γ_{ρ_0} is not real, Weil’s criterion [1, 2] provides $f \in C_c^\infty(\mathbb{R})$ such that, for $g = f * \tilde{f}$, $\sum_{\rho} \hat{g}(\gamma_{\rho}) < 0$; then (3) would give $\int |\hat{f}|^2 d\nu < 0$, impossible for a positive measure. So the intersection is empty. $\square$

Computation 4 (The identity phase). For $a < \frac{1}{2} \log 2 = 0.346574$ no prime power lies inside the cone, and the Weil form at support a is the archimedean and polar part alone. In the odd sector this prime-free form is positive definite for $a < 0.371601$ and indefinite beyond [5, Table 1, row $n = 2$]. Hence, in the odd sector, $\mathcal{P}(a)$ is nonempty for $a < 0.3716$ on the strength of the archimedean term alone, and the first prime power required to keep it nonempty is 2, which enters at $a = 0.3466$ and is needed by $a = 0.3716$. The corresponding entry points and deadlines for 3, 4, 5, 7, 8, 9 are the remaining rows of the same table.

Remark 5 (What the identity fixes and what the primes fix). As $a \downarrow 0$ the test functions in (2) concentrate near $x = 0$, where W is dominated by the singularity of W_{∞} ; this constrains only the large-scale density of ν , which must be that of the Riemann–von Mangoldt law $\frac{1}{2\pi} \log \frac{t}{2\pi}$. Every prime power that enters the cone adds a constraint at the frequency $\log n$, that is, on the fluctuations of ν at the scale $2\pi/\log n$. In this sense the identity prescribes how many zeros there are and the primes prescribe where they are, one frequency at a time, and $\mathcal{P}(a)$ is what remains undetermined at support a .

Remark 6 (The control). Nothing in Proposition 2 uses the Euler product. For the Davenport–Heilbronn function the same construction gives a family $\mathcal{P}_{\text{DH}}(a)$, nonempty for small a by the same archimedean mechanism, and by [5, Remark 12.9] expected to become empty near $a \approx 2.1$, where the horizon $\frac{2\pi}{5}e^{2a}$ reaches the off-line zero at height 85.7. The potential set can therefore contain configurations that are never realized by any zeta function; that it collapses to ν_{ζ} for ζ itself is the Riemann Hypothesis, not a property of the construction.

Remark 7 (Dictionary). The symbolic line “ $1 = A\Omega$ ” reads: at $a = 0^+$ the potential set is everything with the right density; “the first distinction is 2” is Computation 4; “potential defined by the primes” is Definition 1; and the subscript α, ω is the limit $a \rightarrow \infty$ of Proposition 3. The line is not an equation and should not be presented as one; the definition is.

References

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